

The CTC Algorithm

A study guide, stitched from the manim-concepts road

manim-concepts

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How to read this guide

[Phase-2 placeholder: the roadmap figure — the connection graph for this objective drawn as a road, repeated at every transition with progress marked — and the remix map: objective, retrieved sections, and the graph edges justifying their order.] Practice problems close every chapter; worked solutions live in the separate solutions manual, keyed by the same problem numbers. Worked numbers are spliced from the repo’s verification anchors — for instance, the guide’s central worked probability $P(\text{AB} \mid X) = 0.4877$ — never retyped.

Chapter 1

Counting rules

[Phase-2 placeholder.] The four elementary counting rules, retold from the video series: independent stages multiply, ordered selections deplete, unordered selections divide out the order, and partitions divide out the within-group orders. The narrative here will ground every later section: the trellis’s raw path space, the binomial’s columns, and the alignment count all come back to this chapter’s grid.

Practice problems

Problem 1.1. [Calibration placeholder — replaced in phase 2 under the solve gate.] Three lights each show one of three colours, independently. How many distinct displays can the panel show?

Hint: One stage per light; independent stages multiply.